Damage, Weather, or a Broken Sensor? Anytime-Valid Alarm Attribution for Software-Defined AIoT Structural Health Monitoring

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***Abstract*—Data-driven structural health monitoring (SHM) systems usually score each measurement window independently and report every anomaly as possible damage. In a continuously running AIoT deployment this design fails in two ways. First, per-window error rates become hundreds of false alarms per year: even a calibrated 5 % per-window rate means about 440 false alarms per year at one window per hour. Second, alarms caused by environmental variation or by faulty IoT sensors cannot be told apart from real damage. We propose *ShiftGuard*, a monitoring framework with four components: (i) a self-supervised masked-sensor model that predicts each sensor from the rest of the network and temperature; (ii) a mask-and-impute step that isolates any single faulty sensor and fills in lost packets; (iii) temperature-localised conformal p-values that feed three families of conformal e-detectors (CUSUM-type test martingales) for the structure, each sensor and the environment; and (iv) a conformal severity stage that reports damage severity as a prediction set. The alarm comes with an anytime-style guarantee, namely that the expected time to a false alarm is at least a chosen threshold, and each alarm is attributed to structural damage, unprecedented environmental conditions or a specific faulty sensor. The sensing layer is fully software-defined: virtual MQTT sensor nodes replay a bridge model and the Z24 benchmark through a fault injector, which gives exact ground truth for sensor faults that physical deployments cannot provide. Over a simulated monitoring year with a record cold snap, five types of IoT sensor fault and progressive damage, ShiftGuard reduces false-alarm episodes from 315 to about 8 (and false-alarm hours from 347 to 26) compared with a per-window conformal autoencoder baseline, detects moderate damage within 4 h, attributes 94 % of sensor-fault hours to the correct sensor without labelling any of them as damage, and never labels the cold snap as damage. On the real Z24 bridge recordings it keeps false alarms about 40× below the baseline (10 vs. 391 hours) and detects severe damage, but misses the moderate damage stage.**

***Index Terms*—Structural health monitoring, AIoT, conformal prediction, test martingales, e-values, sensor fault isolation, environmental variability, Z24 bridge.**

I. Introduction

Bridges and buildings degrade silently, through ageing, weather and repeated loading, and damage often develops inside materials that look fine from the outside. Manual inspection is periodic, expensive and subjective, so early-stage damage that appears between inspection cycles can go unnoticed. IoT sensing combined with artificial intelligence (AIoT) promises continuous, automated monitoring, and the literature on IoT-based SHM has grown rapidly over the last decade [1]–[3].

Three gaps limit how such systems behave once they run continuously.

Gap 1 – sequential guarantees. SHM models report accuracy or F1-score on a test set, or at best a per-window calibrated error rate obtained with conformal prediction [12]. Monitoring, however, is a *sequential* problem. With N windows per year, a per-window guarantee α compounds into $N\alpha$ false alarms, and operators stop trusting the system.

Gap 2 – attribution. A deviation from the healthy baseline has three common causes: structural damage; environmental and operational variation, for example the jump in Z24's natural frequencies when the deck freezes [18]; and IoT sensor faults such as drift, noise, dropout, spikes and saturation. Prior work treats these separately. Temperature compensation and sensor-fault isolation (e.g. FS-LSTM [19] and stacked-GRU frameworks [20]) are separate pipelines, and none comes with a statistical alarm guarantee.

Gap 3 – validation. Physical deployments cannot provide ground truth for which sensor failed, or when.

The main contributions of this work are:

1) A masked-sensor healthy-behaviour model whose residuals separate *local* inconsistency (a sensor fault) from *regional* inconsistency (damage). Its mask-and-impute step also repairs lost packets.
2) The first use, to our knowledge, of conformal e-detectors (CUSUM-type conformal test martingales) for SHM alarms. The guarantee is on the expected time to a false alarm, and temperature-localised conformal calibration handles the seasonal covariate shift.
3) A three-way attribution rule (damage / environment / sensor i) together with conformal severity sets.
4) A software-defined AIoT testbed (MQTT virtual nodes, edge monitor, dashboard, an interactive browser digital twin and Node-RED alerts) with a fault-injection protocol that gives exact ground truth.

II. Related Work

A. IoT-Based SHM

Surveys and bibliometric studies document the range of IoT sensors and data-driven methods used for civil structures [1]–[3], [5], but they do not propose a deployable alarm logic. AIoT architectures that couple wireless sensing with AI-based interpretation have been validated on aerospace structural testing [4], and edge-computing prototypes run lightweight classifiers such as MobileNetV2 with multi-sensor fusion on laboratory bridges [9].

B. Deep Learning on the Z24 Benchmark

Z24 is the standard benchmark for vibration-based damage detection. WaveNet and MiniRocket feature extractors deployed at the edge reach 99.65 % accuracy on five-class Z24 classification [7]. A WaveNet model trained on UAV-mounted accelerometer data predicts damage severity and location [8]. A conditional GAN generates synthetic Z24 vibration signals for damage detection and digital twinning [10], and a TCN-GAT autoencoder trained only on healthy data detects damage on Z24 and the Old Ada Bridge [11]. A hybrid convolutional autoencoder with deep SVDD achieves an F1-score of 98.46 % on an in-situ truss bridge [6], and transformer-based masked autoencoders have been proposed as foundation models for SHM [17]. All of these methods make per-window decisions and do not attribute alarms to a cause.

C. Sensor Fault vs. Damage

FS-LSTM-based isolation [19] and stacked-GRU isolation in noisy environments [20] distinguish sensor faults from structural damage, as do probabilistic sensor-fault detectors for bridge SHM. These are separate classifiers and assume normal environmental conditions.

D. Environmental Variability

The temperature effect is usually removed by regression, principal component analysis or cointegration before damage detection, in a separate pre-processing step [18].

E. Conformal Prediction and Sequential Testing

Conformal prediction provides distribution-free, finite-sample coverage guarantees [12]. Conformal test martingales extend it to change-point detection [14], e-detectors give a nonparametric framework for sequential change detection [15], and WATCH uses weighted-conformal martingales to monitor deployed AI systems [16]. None of these has been applied to SHM.

III. System Architecture

ShiftGuard runs on a software-defined AIoT stack (Fig. 1).

Virtual sensing layer. Eight accelerometer nodes and one temperature node publish 20 s windows (100 Hz) over MQTT once per hour. The source is either an 8-DOF bridge model (first mode about 4 Hz, with Z24-like stiffening below 0 °C) or Z24 recordings replayed with a temperature warp. A fault injector applies gain, drift, noise, spikes, dropout and clipping faults. A Wokwi ESP32 + DHT22 node can override the temperature.

Edge analytics. ShiftGuard (Section IV) reassembles each window from the MQTT packets, imputes lost packets and updates all ten detectors in a median of 4.9 ms per window (p95 9.6 ms) on a laptop CPU, measured over the live pipeline with the ridge model and including feature extraction.

Application layer. Results are stored in SQLite and shown on a Streamlit dashboard. A browser digital twin connects to the broker over MQTT-over-WebSocket: the deck is animated from the live accelerometer windows, and faults, weather and damage are injected by clicking on the model. A Node-RED flow e-mails damage and sensor-fault alerts (including a severity escalation), while weather alerts only go to the log.

Fault tolerance. Crashed components are restarted by a supervisor and the edge monitor resumes its store. Every MQTT client reconnects with back-off and re-subscribes. Windows are processed in strict sequence order, with late packets imputed, and a restarted sensor network is detected and appended to the history. These behaviours are covered by an automated multi-process test with 20 % packet loss and a sensor-network restart.

IV. Proposed Methodology

A. Features

Each window gives, for each sensor i, a vector of $B = 16$ log band powers $x_i \in \mathbb{R}^B$ (1–30 Hz, Welch). The excitation level is removed by subtracting the median over sensors of each sensor's mean log power, so one faulty channel cannot shift the features of the others.

B. Masked-Sensor Model

A network f_θ is trained only on healthy data to predict each sensor from the others:

$$\hat{x}_i = f_\theta\big(x_{-i},\, e_i,\, \phi(T)\big), \tag{1}$$

where x_{-i} is the window with sensor i masked, e_i is a one-hot sensor index and $\phi(T)$ is a temperature basis that includes

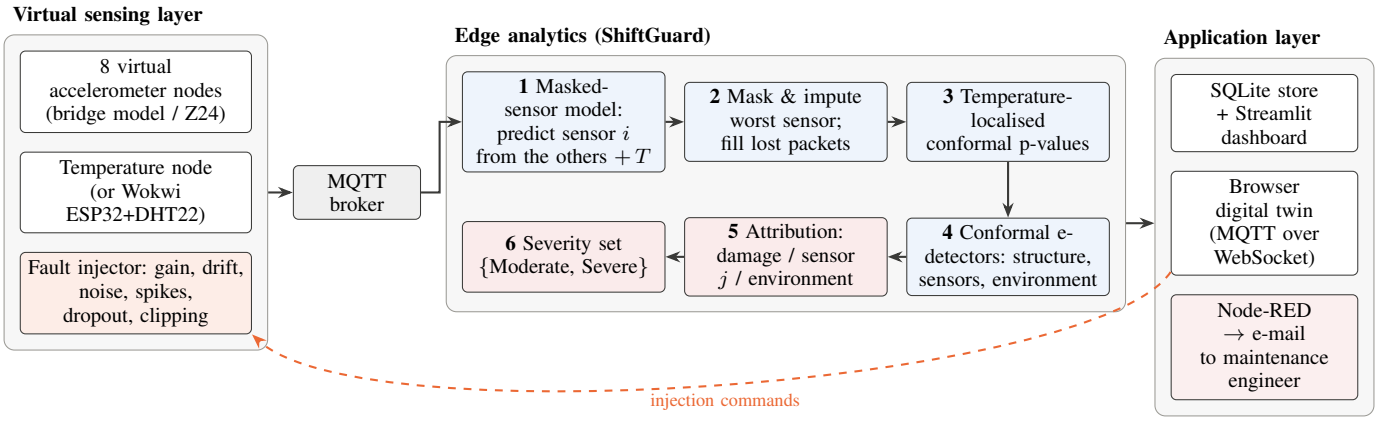


Fig. 1. Software-defined AIoT architecture of ShiftGuard. Virtual sensor nodes publish windows over MQTT; the edge monitor runs the six-step pipeline of Section IV; results feed the dashboard, the digital twin and Node-RED e-mail alerts.

a freezing branch. Training randomly masks sensors, so the model is self-supervised. The sensor score is

$$s_i = \frac{1}{B} \left\| \frac{x_i - \hat{x}_i}{\sigma_i} \right\|^2 . \tag{2}$$

C. Mask-and-Impute

Let $j^* = \arg\max_i s_i$. We replace x_{j^*} by $\hat{x}_{j^*}$, recompute the other sensors' scores s'_i, and define the structure score

$$G = \operatorname*{mean}_{i \neq j^*} s'_i. \tag{3}$$

A faulty sensor then leaves G unchanged, while damage raises it. The per-sensor detectors use the same masked scores: $\tilde{s}_{j^*} = s_{j^*}$ and $\tilde{s}_i = s'_i$ for $i \neq j^*$. Without this, a faulty sensor also inflates the residuals of the neighbours that are predicted from it; their detectors then accumulate evidence, and when the fault clears the alarm is passed on to a neighbour. Lost packets are imputed the same way: features are computed from the received channels only, and each missing sensor is predicted from the others.

D. Temperature-Localised Conformal p-Values

For a score s observed at temperature T, s is ranked among the $k = 400$ calibration scores c recorded at the temperatures closest to T:

$$p = \frac{\#\{c > s\} + U\big(\#\{c = s\} + 1\big)}{k + 1}, \qquad U \sim \mathcal{U}(0, 1). \tag{4}$$

This removes the dependence of model error on temperature, which is a benign covariate shift.

E. Conformal e-Detectors

Each p-value becomes an e-value through a mixture of power calibrators,

$$e = \frac{1}{|E|} \sum_{\varepsilon \in E} \varepsilon \, p^{\varepsilon - 1}, \qquad \mathbb{E}[e] = 1 \text{ under the null.} \tag{5}$$

The detector state is updated as

$$M_t = e_t \, \max(M_{t-1}, 1), \tag{6}$$

and raises an alarm when $M_t \geq c$. If the p-values are i.i.d. uniform (no change), $\mathbb{E}[\tau_{\text{false}}] \geq c$. We use $c = 10^4$ windows, about 14 months at one window per hour. Three families of detectors run in parallel: one for the structure (on G); one for each sensor (on $\tilde{s}_i$, with threshold Nc, a Bonferroni correction); and one for the environment (k-NN novelty of T against the calibration temperatures). The detector state is capped at $10c$ so that it recovers after a condition ends.

F. Attribution

Each window is labelled as follows. ENVIRONMENT: an alarm fires while the environment detector is also in alarm; the conditions have never been seen before, so the alarm is held at low confidence. STRUCTURAL DAMAGE: otherwise, the structure detector is in alarm. SENSOR FAULT j: only a sensor detector is in alarm.

G. Severity

After a damage alarm, a logistic model on $\log s$ outputs a split-conformal prediction set over {Moderate, Severe} with at least 90 % coverage. It is trained on damage at *random* locations, so the test location is never seen in training.

V. EXPERIMENTAL SETUP

A. Monitoring Scenario

The model is trained on one healthy year, with alternate days used for fitting and calibration. It then monitors a second year containing a cold snap 9 °C below the seasonal profile (days 15–22); drift on S3 (days 120–134); noise on S6 (days 200–204); dropout on S1 (days 260–262); spikes on S4 (days 285–288); clipping on S7 (days 300–303); and moderate damage (10 % stiffness loss, days 330–348) followed by severe damage (25 %, days 348–365). One extra healthy-only year checks validity. All results are averaged over 3 random seeds.

TABLE I
SIMULATED MONITORING YEAR, MASKED-SENSOR NEURAL MODEL (MLP). MEAN OF 3 SEEDS.

Method	False-alarm h	False-alarm episodes	Damage delay (h)	Damage recall	Faults correctly attributed	Faults called damage	Cold snap called damage
ShiftGuard (proposed)	**26**	**7.7**	**3.7**	**98.7%**	**93.6%**	**0.0%**	**0.0%**
Per-window AE + conformal 5%	347	314.7	4	67.6%	0.0%	86.1%	36.1%
Per-window AE + conformal 0.1%	7.7	3	326.3	43.8%	0.0%	70.3%	17.3%
Ablation: per-window conformal 5% (no e-detector)	351	328	0	86.7%	0.0%	9.2%	31.9%
Ablation: no sensor masking	40.7	7.3	3.7	98.7%	3.4%	90.9%	0.2%
Ablation: global (not temperature-localised) calibration	106	16.7	7	98.3%	93.9%	0.0%	6.3%
Ablation: no temperature conditioning	28.7	9.3	4.7	98.6%	93.8%	0.1%	0.0%
Ablation: no environment monitor	26.7	7.7	4	99.5%	93.5%	0.0%	33.5%

TABLE II
THE SAME EXPERIMENT WITH THE LINEAR CROSS-SENSOR MODEL (RIDGE). MEAN OF 3 SEEDS.

Method	False-alarm h	False-alarm episodes	Damage delay (h)	Damage recall	Faults correctly attributed	Faults called damage	Cold snap called damage
ShiftGuard (proposed)	**23.3**	**6.7**	**2.7**	**98.8%**	**95.7%**	**1.2%**	**0.0%**
Per-window AE + conformal 5%	347	314.7	4	67.6%	0.0%	86.1%	36.1%
Per-window AE + conformal 0.1%	7.7	3	326.3	43.8%	0.0%	70.3%	17.3%
Ablation: per-window conformal 5% (no e-detector)	377	349.7	0	96.3%	0.0%	18.3%	30.8%
Ablation: no sensor masking	36.3	8.3	2.3	98.8%	1.0%	96.4%	0.0%
Ablation: global (not temperature-localised) calibration	64.3	13	7.3	98.3%	97.3%	0.1%	0.6%
Ablation: no temperature conditioning	26.3	7.7	2.7	98.8%	95.7%	1.3%	0.0%
Ablation: no environment monitor	24	7	3	99.6%	94.9%	2.0%	33.3%

B. Baselines and Ablations

The baseline is an autoencoder-style score (PCA reconstruction error on all features) with a per-window split-conformal threshold at $\alpha = 5\,\%$ and $\alpha = 0.1\,\%$; this is the pipeline from our earlier design. A second baseline uses the ShiftGuard score with a per-window conformal decision. The ablations remove masking, replace localised by global calibration, remove temperature conditioning, and remove the environment monitor.

C. Metrics

We report false-alarm hours and episodes on healthy hours; damage detection delay and recall; the fraction of fault hours attributed to the correct sensor; the fraction of fault and cold-snap hours mislabelled as damage; severity-set coverage; and the empirical false-alarm rate on the healthy-only year.

VI. RESULTS AND DISCUSSION

A. Simulated Monitoring Year

Tables I and II report the simulated year for the neural and the linear variant. Five observations stand out.

Accumulating evidence is what removes false alarms. The per-window decision on the *same* ShiftGuard score gives about 330–350 false-alarm episodes, against 7–8 for the e-detector (Fig. 2).

Masking is what makes fault attribution possible. Without it, 91–96 % of fault hours are called damage, against 0–1 % with masking (Fig. 3).

Localised calibration cuts false-alarm hours by about 75 % (106 $\rightarrow$ 26 h for the MLP, 64 $\rightarrow$ 23 h for ridge) and halves detection delay.

The environment monitor is what stops the cold snap being called damage (33 % $\rightarrow$ 0 %).

The baselines face a dilemma. At $\alpha = 5\,\%$ the baseline detects damage fast but raises 315 false-alarm episodes. At $\alpha = 0.1\,\%$ it misses moderate damage entirely, with a 326 h delay and 44 % recall.

B. Validity of the Guarantee

On the healthy-only year (3 seeds), ShiftGuard raises 3.3 (MLP) and 2.7 (ridge) false-alarm episodes per year, against 390 for the per-window 5 % baseline. The design target is $\mathbb{E}[\text{time to false alarm}] \geq 1.14$ years. Taking into account the per-sensor detectors (a union bound gives $c/2$ overall) and hourly autocorrelation, the realised rate is a few episodes per year: about 100$\times$ fewer than the baseline, but above the nominal bound. Raising c to 10^5 trades about 25 % longer detection delay for fewer alarms.

TABLE III
THE SAME PROTOCOL ON THE REAL Z24 RECORDINGS (RIDGE MODEL, MEAN OF 3 SEEDS). THE LAST ROW USES THE MASKED-SENSOR MLP.

Method	False-alarm h	False-alarm episodes	Damage delay (h)	Damage recall	Faults correctly attributed	Faults called damage	Cold snap called damage
ShiftGuard (proposed)	**10.3**	**5.3**	**494**	**23.5%**	**46.5%**	**0.3%**	**0.0%**
Per-window AE + conformal 5%	391.3	362.7	70.7	10.4%	0.0%	23.0%	16.5%
Per-window AE + conformal 0.1%	12.7	12.3	94.3	1.5%	0.0%	12.6%	2.4%
Ablation: per-window conformal 5% (no e-detector)	367.3	343.3	68	16.0%	0.0%	8.6%	9.7%
Ablation: no sensor masking	16.7	7	37	30.6%	4.8%	41.8%	0.0%
Ablation: global (not temperature-localised) calibration	11.7	6.7	493.3	23.6%	45.7%	2.0%	0.0%
Ablation: no temperature conditioning	10.3	5.3	494.3	23.3%	46.2%	0.3%	0.0%
Ablation: no environment monitor	10.7	5.7	492.7	25.4%	47.9%	0.3%	3.0%
ShiftGuard (MLP)	12.7	5.7	494	27.3%	44.7%	0.0%	0.0%

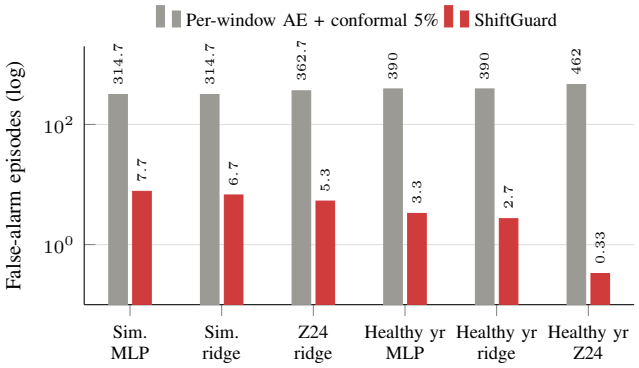


Fig. 2. False-alarm episodes per monitored year: per-window baseline vs. ShiftGuard on the simulated year, on Z24, and on healthy-only validity years (values from Tables I–III).

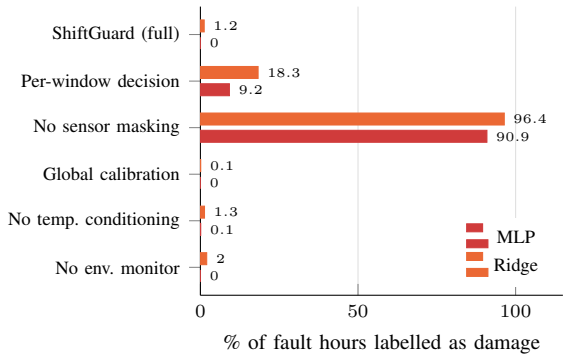


Fig. 3. Ablation: share of sensor-fault hours mislabelled as structural damage (Tables I and II). Removing mask-and-impute turns almost every sensor fault into a false damage alarm.

C. Severity and Localisation

The conformal severity sets reach 100 % (MLP) and 99.7 % (ridge) coverage against a 90 % target, with an average set size of 1.0. During severe damage, the mean per-sensor residual peaks at S5–S6, where the damage was placed. For ridge the residuals are 23.7 and 24.1 against at most 9.3 elsewhere; for the MLP the peak is less sharp (S5 = 11.0, S4 = 9.2).

D. Real Z24 Recordings

The processed Z24 progressive-damage data has 1530 recordings (17 scenarios × 9 sensor setups × 10 segments, 27 channels, 60 s at 100 Hz). Scenarios 1–2 are healthy, scenarios 3–7 and 9–11 are moderate damage, and scenarios 12–17 are severe damage; scenario 8 (the new reference after settlement) is excluded. The cross-sensor model needs a fixed sensor layout. With a single setup there are only 20 healthy recordings, and calibration and test windows then come from different recordings, which breaks exchangeability: ShiftGuard raised 795 false-alarm hours in a 120-day pilot run. We therefore use the 8 channels whose RMS level is as stable *between* setups as *within* a setup (channels 1, 2, 3, 9, 13, 17, 18, 19). These behave like the fixed reference sensors of the Z24 campaign, so all 180 healthy recordings can be used; the selection uses healthy recordings only. Healthy recordings are split 50/25/25 % into fitting, calibration and the monitored stream, and damaged recordings 40/20/40 % into severity training, severity calibration and the stream. Table III reports the results.

False alarms carry over from simulation to real data. ShiftGuard gives 10.3 false-alarm hours (5.3 episodes) against 391 hours (363 episodes) for the per-window baseline. The healthy-only year gives 0.33 false-alarm episodes, against 462 for the baseline.

Weather and sensor faults are still not mistaken for damage. Only 0.3 % of fault hours and 0 % of cold-snap hours are labelled damage, against 23 % and 16.5 % for the baseline.

Damage detection is the weak point on Z24. Recall is 23.5 % and the first damage alarm comes after about 494 h, which is 62 h into the severe stage: the moderate stage (pier settlement, spalling, landslide) is missed on these channels. The baseline is no better (10.4 % recall at $\alpha = 5\,\%$). The early Z24 damage scenarios change the response of the shared reference channels very little.

Fault attribution to the correct sensor drops to 46.5 % (95.7 % in simulation), although faults are almost never called damage. Real recordings have weaker cross-sensor redundancy

than the 8-DOF model, so a fault on one channel is more often blamed on a neighbour.

Severity sets keep their coverage guarantee (92.4 % against a 90 % target) but are less sharp (average set size 1.46, against 1.0 in simulation).

With the masked-sensor MLP instead of ridge, ShiftGuard gives 12.7 false-alarm hours, 27.3 % damage recall and 2.67 false-alarm episodes in the healthy-only year, with 90.4 % severity-set coverage; the conclusions are the same as for ridge.

E. Limitations

Accumulating evidence is what removes false alarms: the ablation that keeps ShiftGuard's score but decides window-by-window is about as noisy as the baseline. Masking is what turns sensor faults from "damage" into correctly localised maintenance tickets, and temperature-localised calibration is what keeps the guarantee valid across seasons. Several limitations remain.

The guarantee is approximate. It assumes exchangeable healthy windows, while hourly windows are autocorrelated; the healthy-only year reports how much this matters in practice.

Single-fault assumption. Mask-and-impute removes one sensor per window, so simultaneous faults would need an iterative version. Lost packets are already imputed iteratively, for any number of missing sensors up to half the network.

Simulated temperature effect. The Z24 temperature effect is simulated, because the progressive-damage test has no temperature channel. Long-term Z24 monitoring data [18] would allow a fully measured evaluation.

Z24 damage sensitivity. On the 8 shared reference channels, moderate Z24 damage is not detected (Table III). The alarm guarantee and the attribution of faults and weather hold on real data, but damage sensitivity depends on sensor placement and features. Using the full setup-specific layouts, or modal features instead of band powers, is future work.

Model choice. The linear (ridge) variant localises damage more sharply, while the neural variant is the self-supervised masked model. Both give the same alarm behaviour, so the framework is model-agnostic.

VII. Conclusion

ShiftGuard reframes SHM alarms as a sequential, attributable decision. It answers *what changed* (the structure, the weather or a sensor) and gives a statistical bound on how often it raises a false alarm. On a simulated monitoring year it cuts false-alarm episodes from 315 to about 8 while detecting moderate damage within 4 h, and on real Z24 recordings it keeps false alarms about 40× below a per-window baseline without mistaking weather or sensor faults for damage. The software-defined AIoT testbed makes this evaluation possible with exact ground truth. Future work will improve damage sensitivity on real structures with setup-specific layouts and modal features, and extend mask-and-impute to simultaneous sensor faults.

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